

Android Studio

1. Calculator App
- I. activity_main.xml

```
1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      android:orientation="vertical"
4      android:padding="20dp"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent">
7
8      <EditText
9          android:id="@+id/num1"
10         android:hint="Enter First Number"
11         android:inputType="numberDecimal"
12         android:layout_width="match_parent"
13         android:layout_height="wrap_content"/>
14
15     <EditText
16         android:id="@+id/num2"
17         android:hint="Enter Second Number"
18         android:inputType="numberDecimal"
19         android:layout_width="match_parent"
20         android:layout_height="wrap_content"
21         android:layout_marginTop="10dp"/>
22
23     <TextView
24         android:id="@+id/result"
25         android:text="Result"
26         android:textSize="20sp"
27         android:layout_marginTop="20dp"
28         android:layout_width="match_parent"
29         android:layout_height="wrap_content"/>
30
31     <Button
32         android:text="Add"
33         android:onClick="add"
34         android:layout_width="match_parent"
35         android:layout_height="wrap_content"/>
36
37     <Button
38         android:text="Subtract"
39         android:onClick="subtract"
40         android:layout_width="match_parent"
41         android:layout_height="wrap_content"/>
42
43     <Button
44         android:text="Multiply"
45         android:onClick="multiply"
46         android:layout_width="match_parent"
47         android:layout_height="wrap_content"/>
48
49     <Button
50         android:text="Divide"
51         android:onClick="divide"
52         android:layout_width="match_parent"
53         android:layout_height="wrap_content"/>
54
55 </LinearLayout>
```

II. MainActivity.java

```
1 package com.devyesh.calculatorapp;
2
3 import android.os.Bundle;
4 import android.view.View;
5 import android.widget.EditText;
6 import android.widget.TextView;
7 import androidx.appcompat.app.AppCompatActivity;
8
9 public class MainActivity extends AppCompatActivity {
10
11     5 usages
12     EditText num1, num2;
13
14     6 usages
15     TextView result;
16
17     @Override
18     protected void onCreate(Bundle savedInstanceState) {
19         super.onCreate(savedInstanceState);
20         setContentView(R.layout.activity_main);
21
22         num1 = findViewById(R.id.num1);
23         num2 = findViewById(R.id.num2);
24         result = findViewById(R.id.result);
25     }
26
27     1 usage
28     public void add(View v) {
29         double a = Double.parseDouble(num1.getText().toString());
30         double b = Double.parseDouble(num2.getText().toString());
31         result.setText("Result: " + (a + b));
32     }
33
34     1 usage
35     public void subtract(View v) {
36         double a = Double.parseDouble(num1.getText().toString());
37         double b = Double.parseDouble(num2.getText().toString());
38         result.setText("Result: " + (a - b));
39     }
40
41     1 usage
42     public void multiply(View v) {
43         double a = Double.parseDouble(num1.getText().toString());
44         double b = Double.parseDouble(num2.getText().toString());
45         result.setText("Result: " + (a * b));
46     }
47
48     1 usage
49     public void divide(View v) {
50         double a = Double.parseDouble(num1.getText().toString());
51         double b = Double.parseDouble(num2.getText().toString());
52
53         if (b == 0) { } else { }
54     }
55 }
```

Output:

2:12

467

983

Result: 1450.0

Add

Subtract

Multiply

Divide

2:12

467

983

Result: -516.0

Add

Subtract

Multiply

Divide

2. Email App

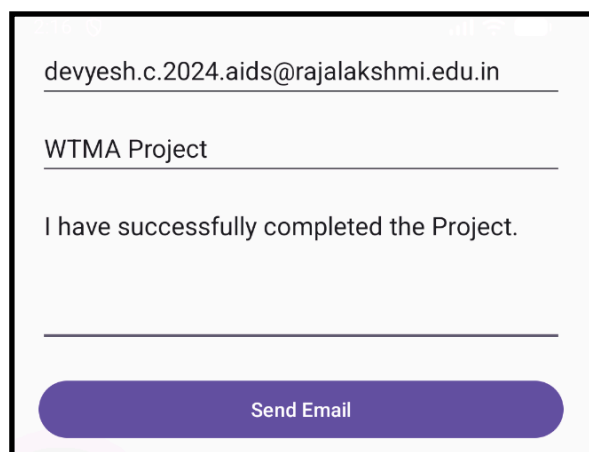
I. activity_main.xml

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3     android:orientation="vertical"
4     android:padding="20dp"
5     android:layout_width="match_parent"
6     android:layout_height="match_parent">
7
8     <EditText
9         android:id="@+id/email"
10        android:hint="Enter Recipient Email"
11        android:inputType="textEmailAddress"
12        android:layout_width="match_parent"
13        android:layout_height="wrap_content"/>
14
15    <EditText
16        android:id="@+id/subject"
17        android:hint="Enter Subject"
18        android:layout_width="match_parent"
19        android:layout_height="wrap_content"
20        android:layout_marginTop="10dp"/>
21
22    <EditText
23        android:id="@+id/message"
24        android:hint="Enter Message"
25        android:layout_width="match_parent"
26        android:layout_height="wrap_content"
27        android:layout_marginTop="10dp"
28        android:minLines="4"
29        android:gravity="top"/>
30
31    <Button
32        android:id="@+id/sendButton"
33        android:text="Send Email"
34        android:layout_width="match_parent"
35        android:layout_height="wrap_content"
36        android:layout_marginTop="20dp"/>
```

II. MainActivity.java

```
1 package com.devyesh.emailapp;
2
3 > import ...
10
11 </> public class MainActivity extends AppCompatActivity {
12
13     2 usages
14     EditText email, subject, message;
15     2 usages
16     Button sendButton;
17
18     @Override
19     protected void onCreate(Bundle savedInstanceState) {
20         super.onCreate(savedInstanceState);
21         setContentView(R.layout.activity_main);
22
23         email = findViewById(R.id.email);
24         subject = findViewById(R.id.subject);
25         message = findViewById(R.id.message);
26         sendButton = findViewById(R.id.sendButton);
27
28         sendButton.setOnClickListener(new View.OnClickListener() {
29             @Override
30             public void onClick(View v) {
31
32                 String recipient = email.getText().toString();
33                 String sub = subject.getText().toString();
34                 String msg = message.getText().toString();
35
36                 Intent intent = new Intent(Intent.ACTION_SEND);
37                 intent.setType("message/rfc822");
38                 intent.putExtra(Intent.EXTRA_EMAIL, new String[]{recipient});
39                 intent.putExtra(Intent.EXTRA_SUBJECT, sub);
40                 intent.putExtra(Intent.EXTRA_TEXT, msg);
41
42                 startActivity(Intent.createChooser(intent, title: "Choose Email Client"));
```

Output:



The screenshot shows the MainActivity app interface. It has a light blue background with a white rounded rectangle containing the form. The form has three text input fields: the first contains 'devyesh.c.2024.aids@rajalakshmi.edu.in', the second contains 'WTMA Project', and the third contains 'I have successfully completed the Project.'. Below the fields is a blue rounded button with the text 'Send Email'.

3. Font Colour Toast App

I. MainActivity.java:

```
1 package com.devyesh.fontcolortoastapp;
2
3 import android.graphics.Color;
4 import android.graphics.Typeface;
5 import android.os.Bundle;
6 import android.view.View;
7 import android.widget.Button;
8 import android.widget.TextView;
9 import android.widget.Toast;
10
11 import androidx.appcompat.app.AppCompatActivity;
12
13 public class MainActivity extends AppCompatActivity {
14
15     16 usages
16     TextView textView;
17
18     2 usages
19     Button button;
20
21     3 usages
22     int styleIndex = 0; // To track current style
23
24     @Override
25     protected void onCreate(Bundle savedInstanceState) {
26         super.onCreate(savedInstanceState);
27         setContentView(R.layout.activity_main);
28
29         textView = findViewById(R.id.textView);
30         button = findViewById(R.id.button);
31
32         button.setOnClickListener(new View.OnClickListener() {
33             @Override
34             public void onClick(View v) {
35
36                 textView.setTypeface(Typeface.DEFAULT, Typeface.NORMAL);
37                 Toast.makeText(context: MainActivity.this, text: "Style 1 Applied", Toast.LENGTH_SHORT).show();
38                 break;
39
40                 case 1:
41                     textView.setTextColor(Color.RED);
42                     textView.setTextSize(26);
43                     textView.setTypeface(Typeface.SERIF, Typeface.BOLD);
44                     Toast.makeText(context: MainActivity.this, text: "Style 2 Applied", Toast.LENGTH_SHORT).show();
45                     break;
46
47                 case 2:
48                     textView.setTextColor(Color.BLUE);
49                     textView.setTextSize(28);
50                     textView.setTypeface(Typeface.MONOSPACE, Typeface.ITALIC);
51                     Toast.makeText(context: MainActivity.this, text: "Style 3 Applied", Toast.LENGTH_SHORT).show();
52                     break;
53
54                 case 3:
55                     textView.setTextColor(Color.GREEN);
56                     textView.setTextSize(30);
57                     textView.setTypeface(Typeface.SANS_SERIF, Typeface.BOLD_ITALIC);
58                     Toast.makeText(context: MainActivity.this, text: "Style 4 Applied", Toast.LENGTH_SHORT).show();
59                     break;
60
61                 case 4:
62                     textView.setTextColor(Color.MAGENTA);
63                     textView.setTextSize(32);
64                     textView.setTypeface(Typeface.DEFAULT_BOLD);
65                     Toast.makeText(context: MainActivity.this, text: "Style 5 Applied", Toast.LENGTH_SHORT).show();
66                     break;
67
68                 case 5:
69                     textView.setTextColor(Color.CYAN);
70                     textView.setTextSize(34);
71                     textView.setTypeface(Typeface.CALLIGRAPH);
72                     Toast.makeText(context: MainActivity.this, text: "Style 6 Applied", Toast.LENGTH_SHORT).show();
73                     break;
74
75                 case 6:
76                     textView.setTextColor(Color.PINK);
77                     textView.setTextSize(36);
78                     textView.setTypeface(Typeface.CASUAL);
79                     Toast.makeText(context: MainActivity.this, text: "Style 7 Applied", Toast.LENGTH_SHORT).show();
80                     break;
81
82                 case 7:
83                     textView.setTextColor(Color.LIGHT_GRAY);
84                     textView.setTextSize(38);
85                     textView.setTypeface(Typeface.SANS_SERIF);
86                     Toast.makeText(context: MainActivity.this, text: "Style 8 Applied", Toast.LENGTH_SHORT).show();
87                     break;
88
89                 case 8:
90                     textView.setTextColor(Color.DARK_GRAY);
91                     textView.setTextSize(40);
92                     textView.setTypeface(Typeface.SANS_SERIF);
93                     Toast.makeText(context: MainActivity.this, text: "Style 9 Applied", Toast.LENGTH_SHORT).show();
94                     break;
95
96                 case 9:
97                     textView.setTextColor(Color.BLACK);
98                     textView.setTextSize(42);
99                     textView.setTypeface(Typeface.SANS_SERIF);
100                    Toast.makeText(context: MainActivity.this, text: "Style 10 Applied", Toast.LENGTH_SHORT).show();
101                    break;
102            }
103        });
104    }
105}
```

II. activity_main.java

```
1  <?xml version="1.0" encoding="utf-8"?>
2  <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3      android:orientation="vertical"
4      android:gravity="center"
5      android:padding="20dp"
6      android:layout_width="match_parent"
7      android:layout_height="match_parent">
8
9      <TextView
10         android:id="@+id/textView"
11         android:text="Hello Android!"
12         android:textSize="24sp"
13         android:textStyle="normal"
14         android:textColor="#000000"
15         android:layout_width="wrap_content"
16         android:layout_height="wrap_content"/>
17
18     <Button
19         android:id="@+id/button"
20         android:text="Change Style"
21         android:layout_marginTop="20dp"
22         android:layout_width="wrap_content"
23         android:layout_height="wrap_content"/>
24
25 </LinearLayout>
```

Output:

